BEU CONNECT

Aryabhatta Knowledge University, Patna

B.Tech 5th Semester Examination, 2017

Time: 03 Hours

Code: 031511 ***Subject: Microprocessor and its Application*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer as directed any seven of the following: **[2x7=14]**

(a) How many general purpose registers are there in 8085 microprocessor?

(b) Program counter is an 8-bit register. (State True or False)

(c) How many address buses are there in 8085 microprocessor?

(d) If sign flag is set result of the arithmetic operation is positive. (State True or False)

(e) How many pins are there in 8085 microprocessor?

(f) What is the addressing mode of the instruction STA 2400H?

(g) LXI H, 2500H is a 3-byte instruction. (State True or False)

(h) How many segment registers are there in 8086 microprocessor?

(i) How many data buses are there in 8086 microprocessor?

(j) Intel 8251 is a programmable counter/interval timer. (State True or False)

Q.2 Solve both questions : **[8+6=14]**

(a) State the functions of the following pins of 8085 microprocessor: ALE, *IO*/overline*M*, S0, S1, HLDA

(b) Discuss flag register of 8085 microprocessor.

Q.3 Solve both questions : **[8+6=14]**

(a) What are the functions of the following instructions of 8085 microprocessor? LXI H, 2400H, RAL, LDA 2500H, IN 01

(b) Explain addressing modes of 8085 microprocessor with suitable examples.

Q.4 Solve both questions : **[9+5=14]**

(a) Draw the timing diagram of MVI A, 05.

(b) Discuss the functions of program counter and stack pointer.

Q.5 Solve both questions : **[5+9=14]**

(a) Write an assembly language program to add two 8-bit numbers and sum is 16 bits.

(b) Write an assembly language program to find largest no. in a data array.

Q.6 Solve both questions : **[6+8=14]**

(a) What are the functions of the following pins of Intel 8257? DRQ0 - DRQ3, A_6– A_7, AEN, ADSTB

(b) Draw and discuss the pin description of Intel 8251.

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

BEU CONNECT

Q.7 Solve both questions : **[6+8=14]**

(a) Write the functions of the following pins of 8086 microprocessor: *BHE*/*S*7, NMI, overline*TEST*, overline*DEN*

(b) Discuss register organization of 8086 microprocessor.

Q.8 Solve both questions : **[6+8=14]**

(a) Write the functions of the following instructions of 8086 microprocessor: MOV AX, DX, ADD AL, 35H, DIV CL, LODSB

(b) Write an assembly language program to add series of 16 bit numbers and sum is 32 bits.

Q.9 Write short notes on the following: **[7x2=14]**

(a) Interfacing to ADC

(b) Data transfer schemes

BEU CONNECT

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified

PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students

GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store